Hydrogenic Bound States from Phase Holonomy Part I

A Derivation within Super Information Theory

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Abstract

In standard atomic physics, hydrogenic binding and spectral quantization follow from the $U(1)$ gauge structure of electromagnetism together with the Schrödinger or Dirac eigenvalue problem. Super Information Theory (SIT) is an effective framework built from a time-density scalar ρ_t and a complex coherence field $\psi = R_{\text{coh}} e^{i\theta}$, and is designed to recover ordinary quantum field theory (QFT) in a constant-background (decohered) limit. In this paper we present a conservative consistency result: in the QFT/decohered limit, the SIT effective action reduces to a locally $U(1)$ gauge-invariant matter–electromagnetic sector (QED), possibly modified by gauge-invariant dressing functions that depend on informational backgrounds. We show that the static field of a localized source obeys the corresponding (dressed) Gauss law, yielding an inverse-square electric field $E_r(r) \propto 1/r^2$ and a Coulomb potential $\phi(r) \propto 1/r$ when the dressings are approximately constant on atomic scales. We then formulate orbital quantization in a gauge-covariant manner using the connection/holonomy language natural to SIT, and recover the Bohr–Sommerfeld limit as a semiclassical expression of global single-valuedness. The full discrete hydrogenic spectrum is thereby understood as that of the recovered Schrödinger/Dirac problem, with SIT providing a compatible geometric interpretation and a falsifiable pathway for small, environment-dependent deviations when time-density or coherence gradients become appreciable.

Relation to complementary work. The analysis presented here adopts a deliberately conservative posture, establishing that Super Information Theory admits a well-defined QFT/decohered limit in which standard electromagnetic binding and hydrogenic quantization are recovered through the usual $U(1)$ gauge structure and eigenvalue problem. A complementary manuscript, *Hydrogenic Bound States from Phase Holonomy Part II*, examines a deeper foundational question—whether inverse-square attraction and discrete bound configurations can arise directly from informational geometry and coherence topology without assuming electromagnetism or quantum boundary conditions as primitive. The two analyses operate at distinct descriptive layers and are not in competition: the present paper secures continuity with established atomic physics, while the companion work explores a possible explanatory origin of that structure at a more fundamental level.

1 Introduction: From Probability to Informational Geometry

The hydrogen atom remains the canonical laboratory for understanding why matter forms stable, discrete structures. In standard atomic physics, stability and quantization are explained within a

locally $U(1)$ gauge-invariant quantum theory: the Coulomb potential arises as the static solution of Maxwell's equations for a localized source, and discrete energy levels arise as eigenvalues of the Schrödinger or Dirac Hamiltonian subject to normalizability and regularity conditions. This framework is extraordinarily predictive.

Super Information Theory (SIT) proposes a deeper effective description in which familiar low-energy fields and interactions are embedded in a theory built from informational structure. In the SIT formalism, two primitive informational fields are introduced: a real time-density scalar $\rho_t(x)$ and a complex coherence field

$$\psi(x) = R_{\mathrm{coh}}(x) e^{i\theta(x)}. \tag{1}$$

The theory admits gauge-invariant matter and gauge sectors that may be dressed by functions of the informational background. A central requirement of viability is that, in an appropriate constant-background (decohered) limit, SIT reduces to ordinary quantum field theory (QFT) (in particular, the $U(1)$ gauge-invariant matter–electromagnetic sector relevant for atomic physics).

The aim of this paper is therefore conservative: we provide a *consistency demonstration* that the hydrogenic binding problem is recovered in the SIT QFT/decohered limit, while clarifying where SIT-specific effects can enter as controlled corrections when informational gradients become non-negligible. In this posture, SIT does not replace the standard eigenvalue description of hydrogen; rather, it supplies a compatible geometric language (connection, curvature, holonomy, global single-valuedness) and an effective-field-theory framework in which small deviations can be parameterized and experimentally constrained.

1.1 Informational fields and the effective low-energy description

SIT assigns an informational energy scale to coherence and time-density through an effective relation of the form

$$E(x) = \zeta\, R_{\mathrm{coh}}(x)\, [\rho_t(x)]^2, \tag{2}$$

with ζ a constant setting the conversion between informational structure and physical energy in the chosen normalization. This relation motivates the interpretation that localized, persistent excitations ("particles" in the low-energy theory) correspond to regions where coherence is non-negligible and stable under the relevant dynamics. In the constant-background limit, one may associate an effective rest-mass scale with localized excitations via $m \sim E/c^2$, but throughout this paper we do not rely on a nonstandard force law obtained directly from gradients of (2); instead we use the recovered gauge-invariant field equations in the SIT QFT limit.

1.2 Electromagnetism as $U(1)$ gauge geometry in SIT

In the SIT framework, the phase structure of ψ motivates a $U(1)$ gauge geometry described by a connection A_μ and curvature $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. In an effective-field-theory formulation consistent with SIT, the electromagnetic sector may appear in a dressed but gauge-invariant form,

$$\mathcal{L}_{\mathrm{EM}} = -\frac{1}{4}\, f_2(\rho_t, |\psi|)\, F_{\mu\nu}F^{\mu\nu} \;-\; J^\mu A_\mu, \tag{3}$$

with J^μ the conserved $U(1)$ matter current and f_2 a nonnegative dressing function. In the decohered/QFT limit relevant for ordinary atomic physics, $f_2(\rho_t, |\psi|)$ is approximately constant across the atomic region, and the action reduces to the usual Maxwell–matter form. In Section 3 we show that the inverse-square Coulomb field then follows from the (dressed) Gauss law, with possible SIT-dependent deviations entering only through weak spatial variation of f_2.

1.3 Quantization: global single-valuedness and holonomy

SIT's gauge-geometric viewpoint also provides a natural language for quantization. The gauge-invariant observable associated with phase transport is the Wilson loop (holonomy) around a closed curve γ,

$$U[\gamma] = \exp\left(\frac{iq}{\hbar} \oint_\gamma A_\mu \, dx^\mu\right). \tag{4}$$

However, physical quantization does not require all Wilson loops to be trivial. Rather, the relevant constraint is that physically admissible quantum states must be globally well defined (single-valued up to an overall phase), which yields a gauge-covariant condition on the accumulated phase of a transported state. In semiclassical language this becomes the Bohr–Sommerfeld quantization rule for canonical momentum. In Section 4 we present this quantization condition in gauge-covariant form and recover the familiar Bohr scalings as a semiclassical limit, while emphasizing that the full hydrogenic spectrum is obtained by solving the recovered Schrödinger/Dirac eigenvalue problem in the SIT QFT limit. Standard results from quantum mechanics, quantum electrodynamics, and gauge theory are cited where they are invoked, and are recovered here only in the explicitly stated QFT/decohered limit.

1.4 Roadmap

The remainder of the paper is organized as follows. Section 3 derives the Coulomb field from the recovered (possibly dressed) $U(1)$ gauge sector and Gauss' law. Section 4 formulates quantization in gauge-covariant holonomy language and recovers the semiclassical Bohr scalings as a limit of the standard hydrogenic problem. Section 5 states explicitly the QFT/decohered limit used throughout and delineates the scope of claims, including the role of possible SIT-specific corrections. We conclude by summarizing the recovery result and outlining a falsifiable experimental pathway for bounding (or detecting) small environment-dependent deviations correlated with informational and coherence structure.

A companion manuscript (Part II) examines the same system from the internal dynamical perspective of Super Information Theory, establishing the necessity of discrete hydrogenic structure.

2 Theoretical Framework: Core SIT Fields and Master Relations

Super Information Theory (SIT) is formulated around the postulate of *Informational Realism*: physical phenomena may be modeled as emergent behavior of an underlying informational substrate. For the purposes of this manuscript, we require only a minimal subset of the SIT framework: the definition of the primitive informational fields, their basic interpretation, and the master energy-density relation used to motivate effective-field-theory couplings. We emphasize that, in the conservative posture adopted here, atomic physics is derived in the *QFT/decohered limit* (Section 5) from the recovered gauge-invariant matter–electromagnetic sector; the master relation below is used to interpret the role of informational backgrounds and to motivate how SIT-dependent corrections may enter through gauge-invariant dressings.

2.1 The two-field basis

SIT introduces two primitive informational fields defined on spacetime:

- **Coherence amplitude (Coherence Ratio) $R_{\mathrm{coh}}(x)$.** This is a dimensionless scalar field constrained to

$$0 \leq R_{\mathrm{coh}}(x) \leq 1,$$

intended to quantify the local degree of coherent order (phase alignment / structured correlation) in the underlying informational substrate. In practice, R_{coh} is treated as the amplitude of a complex coherence field ψ (below), and it provides a scalar measure of how strongly coherent dynamics contribute to the local effective description.

- **Time-density $\rho_t(x)$.** This is a real scalar field with units of inverse time, $[\rho_t] = T^{-1}$, intended to encode the local rate of informational state updates (causal transitions) and thereby set a local dynamical scale. When ρ_t is approximately constant, local dynamics reduce to the familiar constant-rate descriptions of the recovered QFT sector.

It is often convenient to package coherence into a complex field

$$\psi(x) \equiv R_{\mathrm{coh}}(x)\, e^{i\theta(x)}, \tag{5}$$

where $\theta(x)$ is a phase variable whose gauge-geometric interpretation is developed in Sections 3–4.

2.2 The informational energy-density relation

SIT motivates an effective notion of "informational energy density" associated with the local informational background. In the normalization adopted here, this is summarized by the master relation

$$E(x) \;=\; \zeta\, R_{\mathrm{coh}}(x)\, [\rho_t(x)]^2, \tag{6}$$

where ζ is a constant setting the conversion between informational structure and physical energy in the chosen units. In this manuscript $E(x)$ is used as a *background energy-density scale* for the effective theory, not as a mechanical potential to be differentiated into a force.

Dimensional analysis fixes the units of ζ given the interpretation of $E(x)$ as an energy density and $[\rho_t] = T^{-1}$. (The precise normalization of ζ depends on conventions for whether E is treated as an energy density or an energy scale per coarse-graining cell; SIT permits either choice provided it is used consistently.)

2.3 How the master relation enters the effective field theory

The role of (6) in the present paper is twofold:

1. **Interpretation of localized excitations.** When R_{coh} is non-negligible and stable under the relevant dynamics, SIT motivates the interpretation that localized, persistent excitations carry an effective rest-energy scale $E \sim mc^2$ in the constant-background limit. This provides an informational interpretation of "inertia" without altering the recovered QFT dynamics used for atomic structure.

2. **Motivation for controlled, gauge-invariant dressings.** In the SIT effective description, Standard Model and gauge-theory building blocks may be dressed by functions of the informational background, schematically $f_i(\rho_t, |\psi|)\, \mathcal{L}_i$. In the QFT/decohered limit these dressings approach constants, recovering ordinary QFT. When informational gradients become appreciable, the f_i provide a principled and falsifiable way to parameterize small deviations (e.g. renormalizations of effective couplings or masses) while preserving the required gauge symmetries.

In the sections that follow we use this posture explicitly: the Coulomb field is recovered from the $U(1)$ gauge sector and Gauss' law, and orbital quantization is formulated via gauge-covariant single-valuedness in the recovered quantum theory, with SIT-specific contributions entering only as controlled background dressings.

2.4 Emergent mass

A central interpretive element of SIT is that what we call "rest mass" can be associated with stable, localized informational energy stored in the coherence/time-density background. In the constant-background (QFT/decohered) regime, where ordinary relativistic kinematics applies, one may relate an effective rest-energy scale to an effective rest mass via $E \sim mc^2$. Combining this identification with the master energy-density relation (6) motivates the emergent-mass scaling

$$m(x) \;\equiv\; \frac{1}{c^2}\,E(x) \;=\; \frac{\zeta}{c^2}\,R_{\mathrm{coh}}(x)\,[\rho_t(x)]^2. \tag{7}$$

In this manuscript, (7) is used as a *background interpretation* of how localized excitations may acquire an effective inertial scale in SIT, not as a modification of the standard equations of motion in the QFT limit. In particular, atomic binding and quantization are derived from the recovered gauge-invariant matter–electromagnetic sector (Sections 3 and 4). SIT-specific effects enter, if at all, through controlled environment-dependent dressings of effective parameters when R_{coh} and/or ρ_t vary appreciably.

With this posture, "massive" excitations correspond to persistent localized configurations for which R_{coh} is non-negligible and the effective background is approximately stationary. Conversely, "massless" propagation is interpreted as dynamics in which no rest frame exists and the relevant excitation is not associated with a localized rest-energy scale. A detailed dynamical classification of massless versus massive sectors in the full SIT theory is beyond the scope of this paper; here we require only that the QFT/decohered limit reproduces the observed effective masses and couplings of ordinary atomic physics.

3 The Emergence of the Coulomb Force

To derive hydrogenic binding in a way that is faithful to Super Information Theory (SIT), we must identify the long-range interaction between a localized nuclear excitation and a charged matter excitation (the electron, in the QFT limit). In SIT, electromagnetism is not introduced as an additional ontological primitive beyond the theory's gauge structure; it arises from the $U(1)$ geometry associated with the coherence phase and its connection (holonomy) structure. In the appropriate limit, SIT reduces to ordinary QED, and the Coulomb potential follows as the static solution of the dressed Maxwell equations.

3.1 Coherence phase, $U(1)$ structure, and the electromagnetic connection

SIT posits a complex coherence field

$$\psi(x) \equiv R_{\mathrm{coh}}(x)\,e^{i\theta(x)}, \tag{8}$$

with a local $U(1)$ phase symmetry $\psi \mapsto e^{i\alpha(x)}\psi$. Local $U(1)$ invariance requires that derivatives enter through a covariant derivative

$$D_\mu\psi \;\equiv\; \Big(\partial_\mu - i\frac{q}{\hbar}A_\mu\Big)\,\psi, \qquad A_\mu \mapsto A_\mu + \frac{\hbar}{q}\,\partial_\mu\alpha, \tag{9}$$

and defines the gauge curvature (field strength)

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{10}$$

Crucially, A_μ is a genuine connection 1-form and is not globally equal to a pure gradient of θ; only on a simply connected pure-gauge patch (where $F_{\mu\nu} = 0$) may one choose a gauge in which $A_\mu \propto \partial_\mu \theta$.

The gauge-invariant object controlling phase transport is the Wilson line/loop (holonomy)

$$U[\gamma] \;=\; \exp\left(i\frac{q}{\hbar} \int_\gamma A_\mu \, dx^\mu \right), \tag{11}$$

which reduces to $e^{i\Delta\theta}$ only in pure-gauge regions and need not do so globally.

3.2 Electromagnetic sector in the QFT limit of SIT

SIT contains coupling/dressing functions that can make Standard Model parameters weakly dependent on the time-density ρ_t and coherence background. In particular, the electromagnetic kinetic term may be dressed by a nonnegative function $f_2(\rho_t)$:

$$\mathcal{L}_{\rm EM} \;=\; -\frac{1}{4}\, f_2(\rho_t)\, F_{\mu\nu}F^{\mu\nu} \;-\; J^\mu_{\rm EM} A_\mu, \tag{12}$$

where $J^\mu_{\rm EM}$ is the conserved $U(1)$ matter current (Noether current) associated with gauge invariance. In the flat-spacetime, constant-background (decohered/QFT) limit of SIT, the dressings approach constants (often normalized to $f_2 \to 1$), and the action reduces to the standard QFT/QED form with ordinary Maxwell–matter dynamics.

3.3 Dressed Maxwell equation and Gauss law

Varying the action with respect to A_ν yields the Euler–Lagrange (Maxwell) equation

$$\partial_\mu (f_2(\rho_t)\, F^{\mu\nu}) \;=\; J^\nu_{\rm EM}. \tag{13}$$

Taking $\nu = 0$ and writing $E^i \equiv F^{i0}$ gives the dressed Gauss law

$$\nabla \cdot (f_2(\rho_t)\, \mathbf{E}) \;=\; \rho_{\rm EM}, \qquad \rho_{\rm EM} \equiv J^0_{\rm EM}. \tag{14}$$

3.4 Static spherically symmetric solution: emergence of the inverse-square field

Model the nucleus (proton) as a localized, static source of $U(1)$ charge:

$$\rho_{\rm EM}(\mathbf{r}) = Q_p\, \delta^{(3)}(\mathbf{r}), \qquad \mathbf{J} = 0. \tag{15}$$

Assuming spherical symmetry, $\mathbf{E}(\mathbf{r}) = E_r(r)\, \hat{\mathbf{r}}$, and integrating (14) over a sphere of radius r gives

$$\int_{S^2_r} f_2(\rho_t)\, \mathbf{E} \cdot d\mathbf{A} \;=\; 4\pi r^2\, f_2(r)\, E_r(r) \;=\; Q_p, \tag{16}$$

so that for $r > 0$,

$$E_r(r) \;=\; \frac{Q_p}{4\pi\, f_2(r)\, r^2}. \tag{17}$$

If the background varies negligibly across the atomic region (the QFT limit), then $f_2(r) \approx f_2^0$ is approximately constant and the usual inverse-square field follows:

$$E_r(r) \;\approx\; \frac{Q_p}{4\pi f_2^0}\,\frac{1}{r^2}. \tag{18}$$

The electrostatic potential (choosing $\phi(\infty) = 0$) is then

$$\phi(r) \;\approx\; \frac{Q_p}{4\pi f_2^0}\,\frac{1}{r}. \tag{19}$$

A test charge q_e (electron: $q_e = -e$) experiences the Coulomb force

$$F_r(r) \;=\; q_e\, E_r(r) \;\approx\; \frac{q_e Q_p}{4\pi f_2^0}\,\frac{1}{r^2}, \tag{20}$$

which is attractive when $q_e Q_p < 0$.

3.5 Interpretation: what is (and is not) emergent in SIT

In this SIT-consistent derivation, the inverse-square scaling is not obtained by differentiating a postulated radial profile of ρ_t; it follows from (i) $U(1)$ gauge invariance, (ii) the existence of a connection with curvature $F_{\mu\nu}$, and (iii) Gauss's law in three spatial dimensions. The SIT-specific role of ρ_t and coherence enters through dressing functions such as $f_2(\rho_t)$: if f_2 varies measurably in engineered coherence/time-density environments, SIT predicts controlled, environment-dependent renormalizations of electromagnetic response. In the constant-background (QFT) limit, f_2 is effectively constant and standard Coulomb physics is recovered.

4 Topological Stability and Orbital Quantization

With the effective long-range attraction identified, we now address two linked questions: (i) why electronic bound states can be stable (i.e. do not classically "spiral into" the nucleus), and (ii) why discrete quantum numbers arise. In Super Information Theory (SIT), both issues must be consistent with the explicit recovery of the standard $U(1)$ gauge-invariant quantum dynamics (Dirac–Maxwell / Schrödinger) in the constant-background (decohered) limit, while allowing for controlled corrections when informational fields such as ρ_t and $R_{\rm coh}$ vary appreciably.

4.1 Coherence conservation and the meaning of stability

SIT assigns physical significance to the complex coherence field

$$\psi(x) = R_{\rm coh}(x)\, e^{i\theta(x)}. \tag{21}$$

When the effective action depends only on $|\psi|$, global phase invariance $\psi \mapsto e^{i\alpha}\psi$ implies a conserved Noether current (the coherence current),

$$J^\mu_{\rm coh} = \frac{i\kappa_c}{2}\left(\psi\nabla^\mu\psi^* - \psi^*\nabla^\mu\psi\right), \qquad \nabla_\mu J^\mu_{\rm coh} = 0. \tag{22}$$

This conservation law expresses the redistribution and transport of coherence under closed dynamics. It does not, by itself, replace the standard quantum-mechanical explanation of atomic stability.

In the constant-background limit where SIT reduces to ordinary gauge-invariant quantum theory, a bound electron in hydrogen is described by an energy eigenstate of the appropriate Hamiltonian. Such stationary states have time-independent densities and do not radiate continuously, in contrast to a classical accelerating point charge. Radiation arises from non-stationary dynamics, such as transitions between eigenstates. Recovering this behavior is therefore a necessary consistency condition for SIT, rather than an additional postulate.

4.2 Holonomy and gauge-covariant single-valuedness

In the SIT framework, electromagnetism is encoded geometrically by a $U(1)$ connection A_μ with curvature $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The gauge-invariant observable associated with phase transport around a closed curve γ is the Wilson loop (holonomy),

$$U[\gamma] = \exp\left(\frac{iq}{\hbar} \oint_\gamma A_\mu \, dx^\mu\right). \tag{23}$$

The potential A_μ is a genuine connection 1-form and is not, in general, equal to a global gradient of the coherence phase; writing $A \propto d\theta$ is valid only on pure-gauge patches where the curvature vanishes.

Physical quantization does not require the Wilson loop itself to be trivial. Rather, the requirement is that physically admissible quantum states be globally well defined: after parallel transport around a closed loop, the state must return to the same ray. This yields the gauge-covariant single-valuedness condition

$$\exp\left(i \oint_\gamma (\nabla\theta - \frac{q}{\hbar}A) \cdot d\ell\right) = 1, \tag{24}$$

or equivalently,

$$\oint_\gamma (\nabla\theta - \frac{q}{\hbar}A) \cdot d\ell = 2\pi n, \qquad n \in \mathbb{Z}. \tag{25}$$

This is the appropriate topological quantization condition in gauge-covariant form.

4.3 Semiclassical orbital quantization

In the semiclassical (Bohr–Sommerfeld) limit, (25) reduces to quantization of the canonical momentum,

$$\oint_\gamma \mathbf{p}_{\text{can}} \cdot d\ell = 2\pi n\hbar, \qquad \mathbf{p}_{\text{can}} = m\mathbf{v} + q\mathbf{A}. \tag{26}$$

For a purely electrostatic central field and a gauge in which $\mathbf{A} = 0$ along the orbit, this becomes

$$\oint_\gamma m\mathbf{v} \cdot d\ell = 2\pi m v r = 2\pi n\hbar, \tag{27}$$

which implies the familiar angular-momentum quantization

$$L = mvr = n\hbar. \tag{28}$$

If the effective central interaction is Coulombic, $V_{\text{eff}}(r) = -k_{\text{eff}}/r$, the circular-orbit balance condition $mv^2/r = k_{\text{eff}}/r^2$ together with $L = n\hbar$ yields

$$r_n = \frac{n^2\hbar^2}{mk_{\text{eff}}}, \qquad E_n = -\frac{mk_{\text{eff}}^2}{2\hbar^2}\frac{1}{n^2}. \tag{29}$$

These relations reproduce the Bohr scalings and the Rydberg $1/n^2$ spectrum as a semiclassical approximation. A complete description of hydrogenic structure, including the full set of quantum numbers and relativistic corrections, follows from solving the corresponding Schrödinger or Dirac eigenvalue problem in the recovered gauge-invariant sector.

4.4 Scope of SIT-specific corrections

The distinctive contribution of SIT in this context is not a replacement of quantum boundary conditions, but the possibility that informational fields such as ρ_t and the coherence background enter as controlled dressings of effective low-energy parameters while preserving local $U(1)$ gauge invariance. In ordinary atomic conditions, these dressings are effectively constant and the standard hydrogen spectrum is recovered. When coherence or time-density gradients become significant, the kinematic quantization conditions above remain intact, while effective parameters (e.g. couplings or inertial scales) may acquire small, environment-dependent shifts that are, in principle, experimentally testable.

5 Scope and Recovery of Standard Physics

This paper is intentionally conservative in scope. Its purpose is to show how the standard hydrogenic binding problem is recovered *within* the Super Information Theory (SIT) framework when one takes the appropriate low-gradient, decoherence-dominated limit. We therefore make explicit (i) the limiting assumptions under which SIT reduces to ordinary quantum field theory (QFT), (ii) what standard structures are recovered in that limit, and (iii) what is *not* claimed or derived here.

Where the QFT/decohered limit is established. This manuscript does not re-derive the existence of a quantum-field-theory sector from the full SIT dynamics; it works *within* a regime whose reduction is established in the main SIT development. In particular, Appendix E.2 ("Quantum Field Theory Limit: Flat Spacetime and Decohered Fields") shows that when $g_{\mu\nu} \to \eta_{\mu\nu}$ and the informational fields approach constant backgrounds with negligible gradients ($\partial_\mu \rho_t \simeq 0$, $\partial_\mu \psi \simeq 0$), the additional SIT degrees of freedom become non-dynamical and the action collapses to the usual Minkowski-space QFT form,

$$S_{\mathrm{QFT\,lim}} = \int d^4x \, (\mathcal{L}_{\mathrm{matter}} + \mathcal{L}_{\mathrm{EM}}). \tag{30}$$

(See also Sec. 9.2 for the canonical operator structure and the operator/path-integral/classical correspondence.) The present paper therefore takes this limit as an established sector and focuses on the recovered $U(1)$ matter–electromagnetic subtheory relevant for hydrogen.

5.1 The QFT / Decohered Limit of SIT

SIT is formulated in terms of two primary informational fields: the real time-density scalar $\rho_t(x)$ and the complex coherence field $\psi(x) = R_{\mathrm{coh}}(x)e^{i\theta(x)}$, together with gauge and matter sectors that may be dressed by functions of $(\rho_t, R_{\mathrm{coh}})$. The *QFT / decohered limit* used throughout this paper is defined by the following assumptions (cf. the "Quantum Field Theory Limit" statement in the SIT formal development):

- **Flat spacetime:** $g_{\mu\nu} \to \eta_{\mu\nu}$ (gravity decoupled for atomic scales).

- **Constant (or negligibly fluctuating) informational backgrounds:**

$$\rho_t(x) \approx \rho_{t0}, \qquad R_{\rm coh}(x) \approx R_0, \qquad \partial_\mu \rho_t \approx 0, \quad \partial_\mu R_{\rm coh} \approx 0.$$

- **Frozen dressings:** any SIT dressing functions (e.g. $f_1(\rho_t, R_{\rm coh})$, $f_2(\rho_t, R_{\rm coh})$) reduce to constants (often normalized so that $f_i \to 1$), so that matter and electromagnetic fields are effectively uncoupled from $(\rho_t, R_{\rm coh})$ at leading order.

Under these assumptions, the SIT action reduces to the standard Minkowski-space QFT action

$$S_{\text{QFT limit}} = \int d^4x \, (\mathcal{L}_{\rm matter} + \mathcal{L}_{\rm EM}), \tag{31}$$

with canonical operator structure and the usual equal-time commutation relations for the QFT degrees of freedom. In other words, the QFT limit is not an approximation in which one *hopes* to recover known physics; it is a designated regime of the SIT effective field theory in which the additional informational fields become non-dynamical and standard QFT is recovered.

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$$S_{\text{QFT lim}} = \int d^4x \, (\mathcal{L}_{\rm matter} + \mathcal{L}_{\rm EM}). \tag{32}$$

(See also Sec. 9.2 for the canonical operator structure and the operator/path-integral/classical correspondence.) The present paper therefore takes this limit as an established sector and focuses on the recovered $U(1)$ matter–electromagnetic subtheory relevant for hydrogen.

5.2 What Is Recovered Exactly

In the QFT / decohered limit defined above, the theory reproduces the standard gauge-invariant quantum dynamics required for atomic physics. In particular, the following structures are recovered (at leading order in small informational fluctuations):

- $U(1)$ **gauge structure and Maxwell dynamics:** the electromagnetic sector is governed by a $U(1)$ connection A_μ with curvature $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and the Maxwell equations follow from the variational principle applied to $\mathcal{L}_{\rm EM}$.

- **Minimal coupling in the matter sector:** matter couples to the $U(1)$ connection via the usual covariant derivative $D_\mu = \partial_\mu - i(q/\hbar)A_\mu$, ensuring local gauge invariance.

- **Electrostatics and the Coulomb potential:** in the static, spherically symmetric sector sourced by a localized nucleus, Gauss' law yields the inverse-square electric field and the Coulomb potential $\phi(r) \propto 1/r$ (Section 3).

- **Hydrogenic bound states:** the discrete spectrum and quantum numbers arise from solving the corresponding Schrödinger or Dirac eigenvalue problem in the recovered gauge-invariant theory. The semiclassical Bohr–Sommerfeld limit of this result is summarized in Section 4.

SIT's distinctive content in this regime is therefore *not* a replacement of standard atomic physics, but the statement that departures from exact QFT behavior can be organized as controlled effective-field-theory corrections when ρ_t and/or R_{coh} fluctuate weakly about their background values. In that case, their leading effects appear as small renormalizations of effective parameters (e.g. coupling constants and effective mass terms), which can be parametrized and bounded experimentally (e.g. by precision spectroscopy and metrology).

5.3 What Is Not Claimed Here

To avoid overstatement, we list explicitly what this paper does *not* claim:

- **No replacement of quantum mechanics or QED:** we do not propose new postulates that supplant the standard Hilbert-space framework, canonical commutation relations, or local $U(1)$ gauge invariance in the regime where SIT recovers QFT.
- **No "entropic" derivation of electromagnetism:** we do not claim that the Coulomb force arises as a mechanical gradient of a scalar informational energy density. In this paper, Coulomb behavior is recovered from the $U(1)$ gauge sector and Gauss' law in the QFT limit, with possible SIT-specific corrections entering only through gauge-invariant dressing functions.
- **No claim that $U[\gamma] = 1$ is an ontological "existence condition":** holonomy is used in its standard gauge-theoretic sense (parallel transport in a $U(1)$ bundle). Orbital quantization is formulated via gauge-covariant single-valuedness / Bohr–Sommerfeld conditions in the semiclassical limit (Section 4), not as a universal requirement that all Wilson loops be trivial.
- **No full spectroscopy program in this manuscript:** we do not attempt to derive fine structure, hyperfine structure, Lamb shift, radiative widths, or selection rules from first principles here. Those effects arise in the recovered Dirac–Maxwell/QED sector (and, where applicable, from standard loop corrections), with any SIT deviations expected to be small in ordinary laboratory conditions.
- **No claim about numerical parameter values:** this paper does not fix the absolute values of particle masses or couplings from SIT alone. We assume the QFT-limit parameters match known values by construction (as required for viability), and focus only on the structural recovery of the hydrogenic problem and the form of controlled deviations.

Within this carefully delimited scope, the results of Sections 3 and 4 should be read as a consistency demonstration: the SIT effective theory admits a regime in which standard electromagnetic binding and hydrogenic quantization are recovered, while leaving open a falsifiable pathway for small, environment-dependent deviations in regimes where coherence or time-density gradients become appreciable.

5.4 Radiation and Transitions in the SIT Framework

A recurring conceptual concern for any orbital model is the classical prediction that an accelerating charge radiates and loses energy. In the recovered QFT (Dirac–Maxwell / Schrödinger) limit of SIT, this concern is resolved in the standard way: hydrogenic bound states are *stationary* energy eigenstates, and stationary states do not correspond to classical time-dependent point-particle trajectories. Consequently, they do not radiate continuously as classical orbits would. Observable radiation is associated with *non-stationary* dynamics, most notably transitions between eigenstates, and is computed in the usual gauge-invariant framework (e.g. via coupling of the matter current to the quantized electromagnetic field).

Within SIT, the additional informational fields ρ_t and $\psi = R_{\text{coh}} e^{i\theta}$ can affect radiative phenomena only through gauge-invariant modifications of the effective low-energy description. In practice, this means:

- **Stationary stability:** In the constant-background limit, the standard stationary-state argument applies unchanged. SIT does not introduce a new mechanism that forces an electron to "spiral in" nor a new principle that forbids radiation; instead, stability is inherited from the existence of bound eigenstates in the recovered quantum theory.
- **Transitions:** Emission and absorption correspond to transitions between stationary eigenstates driven by perturbations (external fields, collisions) or by interaction with the quantized electromagnetic field (spontaneous emission). In the SIT QFT limit, these processes are described by the same interaction structure as in QED.
- **Possible SIT-specific corrections:** When ρ_t and/or R_{coh} vary appreciably across the interaction region, SIT permits small environment-dependent shifts of effective parameters (e.g. couplings or masses via dressing functions). Such shifts can modify transition frequencies and radiative rates at subleading order, while preserving local $U(1)$ gauge invariance and the standard selection-rule structure at leading order.

In this manuscript we do not attempt a first-principles calculation of radiative line strengths, lifetimes, or spontaneous emission rates from the full SIT dynamics. Instead, we adopt the conservative posture that, in ordinary laboratory conditions, SIT reproduces the standard QED predictions for hydrogen spectroscopy and radiative transitions, with any SIT-dependent deviations expected to enter as small perturbative corrections controlled by the magnitude of informational gradients and coherence-background dressings.

5.5 What SIT adds beyond textbook QED

While Sections 5.1–5.3 delineate the regime in which SIT reduces to ordinary quantum field theory, it is equally important to state explicitly what SIT adds in principle, even when such effects are parametrically small in practice. The purpose of this subsection is to identify the concrete ways in which SIT differs structurally from textbook QED, without assuming that these differences are large in ordinary laboratory conditions.

1. Informational background fields as physical structure. In standard QED, spacetime and coupling constants are treated as fixed background data. In SIT, the time-density $\rho_t(x)$ and coherence amplitude $R_{\text{coh}}(x)$ are physical fields that define an informational background on which the effective low-energy theory is built. In the QFT/decohered limit these fields are approximately

constant and invisible, but in principle they may vary spatially or temporally. This provides a well-defined mechanism for environment-dependent modifications of effective physics that does not require abandoning local Lorentz invariance or gauge symmetry.

2. Gauge-invariant dressing of the electromagnetic and matter sectors. SIT permits the Standard Model building blocks to appear multiplied by gauge-invariant dressing functions of the informational background, schematically

$$\mathcal{L}_{\text{eff}} = \sum_i f_i(\rho_t, |\psi|)\, \mathcal{L}_i^{\text{SM}}.$$

In the electromagnetic sector this takes the form $f_2(\rho_t, |\psi|)F_{\mu\nu}F^{\mu\nu}$, while in the matter sector it may appear as effective mass or coupling renormalizations. In the constant-background limit $f_i \to$ const and textbook QED is recovered exactly; when informational gradients are present, SIT predicts controlled deviations that can be treated within an effective-field-theory expansion.

3. Geometric interpretation of quantization and phase structure. Textbook QED treats quantization primarily as a spectral property of operators on Hilbert space. SIT provides a complementary geometric interpretation: quantization conditions may be expressed in terms of global single-valuedness of sections in a $U(1)$ bundle, holonomy, and phase transport. This does not alter the recovered eigenvalue problem, but it clarifies why integer quantum numbers naturally arise and how semiclassical limits connect to global topology and gauge geometry.

4. Informational interpretation of inertia and rest mass. In standard QED, particle masses are external parameters fixed by experiment. In SIT, rest mass admits an informational interpretation through the background relation $m \sim \zeta R_{\text{coh}}\rho_t^2/c^2$ in the constant-background limit. This does not modify the equations of motion in the recovered QFT sector, but it suggests a framework in which variations of coherence or time-density could, in principle, lead to small, environment-dependent shifts in effective inertial scales.

5. A falsifiable extension rather than a replacement. Taken together, these features position SIT not as a replacement for QED in its domain of validity, but as an extension that embeds QED within a broader informational framework. The predictive content of SIT in the atomic domain lies in the possibility of small, correlated deviations from textbook behavior when coherence or time-density gradients are engineered or encountered in extreme environments. Such deviations can be parameterized, bounded, and in principle detected, providing a clear empirical pathway without undermining the extraordinary success of standard quantum electrodynamics.

5.6 A worked parameterization for $f_2(\rho_t, R_{\text{coh}})$

A natural critique of the recovery analysis above is that the informational dressing functions $f_i(\rho_t, R_{\text{coh}})$ are treated as approximately constant. This is intentional for the purpose of establishing the QFT/decohered limit, but it is nevertheless useful to exhibit an explicit nontrivial parameterization to show how SIT-dependent effects would enter and what they would predict.

Operationally, we identify the coherence amplitude as

$$R_{\text{coh}}(x) \equiv |\psi(x)|, \qquad 0 \le R_{\text{coh}} \le 1, \tag{33}$$

with normalization fixed by the SIT background. We do not attempt a microscopic derivation of $R_{\rm coh}$ here; it is treated as a low-energy order parameter.

Expanding about the constant-background regime $(\rho_t, R_{\rm coh}) = (\rho_{t0}, R_0)$, the leading gauge-invariant dressing of the electromagnetic kinetic term may be written as an effective-field-theory expansion,

$$f_2(\rho_t, R_{\rm coh}) = 1 + \epsilon\,\frac{\rho_t - \rho_{t0}}{\rho_{t0}} + \eta\,\frac{R_{\rm coh}^2 - R_0^2}{R_0^2} + \mathcal{O}\left(\frac{\nabla^2}{\Lambda^2}\right), \tag{34}$$

where ϵ and η are dimensionless Wilson coefficients and Λ is the EFT cutoff scale controlling higher-derivative corrections.

For a static, spherically symmetric source, the dressed Gauss law then yields

$$E_r(r) = \frac{Q}{4\pi r^2 f_2(r)} \;\approx\; \frac{Q}{4\pi r^2}[1 - \delta f_2(r)], \qquad \delta f_2(r) \equiv f_2(r) - 1, \tag{35}$$

and hence a modified Coulomb potential

$$\phi(r) \approx \frac{Q}{4\pi r}[1 - \delta f_2(r)]. \tag{36}$$

In hydrogenic bound states this induces perturbative energy shifts whose leading fractional magnitude scales as

$$\frac{\delta E_n}{E_n} \sim -2\,\frac{\delta f_2}{f_2}, \tag{37}$$

up to expectation-value weighting over the bound-state wavefunctions. Precision spectroscopy therefore constrains δf_2 (and hence informational-gradient effects) to be extremely small in ordinary laboratory conditions.

This example illustrates the intended role of $R_{\rm coh}$ and f_2 in SIT: they do not replace the recovered QED dynamics, but provide a controlled, gauge-invariant channel through which environment-dependent deviations may be parameterized, bounded, and, in principle, detected. A first-principles computation of $f_2(\rho_t, R_{\rm coh})$ from full SIT dynamics is left for future work.

6 Discussion

The preceding sections were deliberately scoped to a conservative goal: to demonstrate that SIT admits a regime in which the standard hydrogenic binding problem is recovered, while clarifying where SIT-specific structure may enter as controlled corrections. In this Discussion we summarize what has (and has not) been established, and we outline the most defensible interpretive and experimental implications.

6.1 Interpretation: geometry, topology, and the standard eigenvalue picture

In standard quantum mechanics, discrete hydrogenic spectra arise as eigenvalues of a self-adjoint Hamiltonian acting on a Hilbert space, with normalizability and regularity conditions selecting a discrete set of bound states. In the QFT/decohered limit, SIT is constructed to recover this framework: the Dirac–Maxwell (or Schrödinger) eigenvalue problem remains the correct description of stationary bound states, and the principal quantum number n retains its standard meaning as an energy-level label.

SIT nevertheless offers a complementary *geometric* interpretation of quantization. Because the matter sector is locally $U(1)$ gauge invariant, physically admissible states must be globally well-defined sections of the associated bundle. In semiclassical language this appears as a gauge-covariant single-valuedness condition for the accumulated phase,

$$\oint_\gamma (\nabla\theta - \frac{q}{\hbar}A) \cdot d\ell = 2\pi n, \qquad n \in \mathbb{Z},$$

which reduces to the Bohr–Sommerfeld rule in appropriate limits. Thus, SIT can be read as highlighting that certain integer labels in the spectrum are consistent with (and, in the semiclassical regime, may be expressed by) topological and geometric constraints associated with phase transport, holonomy, and global single-valuedness. Importantly, this interpretation does not supplant the eigenvalue problem in the regime where SIT recovers QFT; it provides an additional geometric lens through which the same discrete structure may be understood.

6.2 "Forbidden regions" and what SIT does (and does not) imply

It is tempting to reinterpret the classically forbidden regions of standard quantum mechanics as "zones of non-existence." However, such a claim would require a dynamical computation showing that the coherence amplitude R_{coh} must collapse to zero in those regions and that the result is consistent with gauge invariance, unitarity, and energy accounting. No such derivation is presented here, and we do not assume it.

Within the scope of this paper, the correct and conservative statement is the following: in stationary hydrogenic eigenstates the electron is not localized on a classical orbit, and its radial probability density is determined by the relevant eigenfunction. Regions of suppressed probability (including nodes) arise from the structure of the wavefunction; they do not require the electron to lose its rest mass or "dissolve" into the vacuum.

What SIT can plausibly add, consistent with the effective-field-theory posture adopted above, is a mechanism by which strong gradients or engineered coherence environments modify effective parameters through dressing functions (e.g. small shifts in couplings, effective masses, or response functions), thereby perturbing bound-state wavefunctions, transition frequencies, and rates. Establishing any stronger "exclusion" mechanism would require a dedicated dynamical model of R_{coh} backreaction and its coupling to the matter and gauge sectors, which lies beyond the present manuscript.

6.3 Macroscopic implications and a falsifiable pathway

SIT's informational energy-density relation suggests that coherence can, in principle, enter the effective stress-energy budget through the dependence of the low-energy theory on the coherence background. This raises a natural question: can macroscopic coherent systems exhibit environment-dependent gravitational or inertial effects beyond those expected from their ordinary energy content?

Any such effect must be formulated carefully. In relativistic physics, gravitational sourcing is determined by the full stress-energy tensor, and any enhancement claim must either (i) correspond to a change in total energy (including internal and field contributions), or (ii) introduce a new coupling channel consistent with covariance and existing experimental constraints. The conservative, reviewer-proof position is therefore:

- SIT motivates the study of *coherence-dependent* corrections to effective stress-energy through well-defined dressing functions in the unified action.

- In ordinary laboratory settings, any such corrections must be small, or they would already be excluded by precision equivalence-principle tests and gravimetry.
- The appropriate "smoking gun" program is therefore not an a priori claim of large anomalous gravity from coherence, but a parameterized search for small deviations correlated with controlled changes in coherence (e.g. in macroscopic quantum states), with explicit bounds.

A concrete experimental direction is precision differential measurement: compare otherwise matched systems in coherent versus incoherent states while controlling total mass–energy and systematics. Within the SIT effective-field-theory posture adopted here, any observable deviation should be expressible as a small, dimensionless correction governed by the magnitude of informational gradients and coherence dressings, and can therefore be bounded even in the absence of a fully microscopic calculation.

6.4 Summary

In summary, this paper establishes a consistency result: SIT admits a controlled regime in which the standard Coulomb interaction and hydrogenic quantization are recovered, while providing a geometric vocabulary (connection, holonomy, global single-valuedness) that naturally accompanies the recovered $U(1)$ gauge sector. Claims stronger than this—such as dynamical "non-existence" in forbidden regions or large macroscopic gravitational anomalies—require additional dynamical input and dedicated confrontation with existing experimental constraints, and are left for future work.

7 Conclusion

This manuscript has shown how the standard hydrogenic binding problem can be embedded within the Super Information Theory (SIT) framework in a controlled and conservative manner. Rather than replacing quantum electrodynamics (QED) in its domain of validity, we made explicit the *QFT/decohered limit* of SIT in which the effective low-energy theory reduces to the familiar gauge-invariant matter–electromagnetic dynamics, and we used that recovered structure to obtain the inverse-square electrostatic field and the discrete bound-state spectrum. Within this regime, SIT provides a coherent geometric vocabulary—in terms of coherence phase, connection, curvature, and holonomy—while leaving open a falsifiable pathway for small deviations when informational backgrounds become nontrivial.

7.1 Summary of results

1. **Coulomb interaction from the recovered $U(1)$ gauge sector.** We formulated the electromagnetic dynamics in a gauge-invariant manner and showed that the inverse-square electrostatic field follows from Gauss' law for a localized source in three spatial dimensions. In SIT, informational fields such as ρ_t and the coherence background may enter this sector only through gauge-invariant dressing functions (e.g. a multiplier of the $F_{\mu\nu}F^{\mu\nu}$ term). In the constant-background limit relevant to ordinary atomic physics, these dressings are effectively constant and standard Coulomb behavior is recovered.

2. **Quantization as gauge-covariant single-valuedness (and the hydrogen spectrum).** We expressed the quantization condition in a gauge-covariant form using the connection/holonomy language natural to SIT. In the semiclassical limit this reduces to the familiar Bohr–Sommerfeld

quantization of canonical momentum and reproduces the standard Bohr scalings, including the Rydberg $1/n^2$ dependence. More generally, in the recovered QFT limit, the discrete hydrogenic spectrum is obtained by solving the corresponding Schrödinger or Dirac eigenvalue problem in the standard gauge-invariant theory, with SIT providing a geometric interpretation compatible with that framework.

7.2 Outlook and falsifiable extensions

The principal opportunity for SIT to add predictive content in the hydrogenic context is through small, controlled departures from the constant-background regime. If ρ_t and/or R_{coh} vary measurably across the atomic region or within engineered coherence environments, SIT permits environment-dependent dressing of effective low-energy parameters (couplings, effective masses, response functions) while preserving local $U(1)$ gauge invariance. Such effects can be parameterized within an effective-field-theory expansion and confronted with precision spectroscopy and metrology, providing a direct route to empirical constraints.

In this sense, the present work should be read as a consistency demonstration with a clear experimental pathway: SIT admits a regime that reproduces the established electromagnetic binding and hydrogenic quantization of standard physics, and it suggests where and how to search for small deviations correlated with informational and coherence structure in regimes beyond the constant-background limit.

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